

**Topic: Secure Peer-to-Peer Payment System on Android-Based Mobile Devices via
Vibration and Acceleration**

Final Year Project Plan

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CS4514 – Project

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1. Motivation & Background Information

With the increasing use of smartphones and the growth of users in modern society, Near Field Communication (NFC), Wi-fi, and Bluetooth have become popular and standard protocols for short-range wireless communication technologies in mobile phones, particularly in peer-to-peer (P2P) mobile payment systems (Guo et al., 2005). While reliable, this radio frequency (RF)-based approach requires specialized hardware and can be vulnerable to specific eavesdropping and relay attacks (Maatallaoui et al., 2023).

This has encouraged research for an alternative communication channel for P2P data transmission. Some prior work, like using the High Frequency Acoustic Wave channel (Li et al., 2020) and the Visible Light channel (Schmid et al., 2013) for P2P communication. However, mobile phone device vibration and accelerometer sensing have been proposed as a communication channel for data or message transfer and mobile device pairing [Hwang et al., 2014; Yonezawa et al., 2014]. However, the above-mentioned approaches have primarily focused on establishing a basic communication connection and transmitting simple strings only for experimental purposes. They often assume that the channel itself is a trusted and safe. Implementing a secure protocol for the physical vibration channel remains largely unexplored, particularly in financial applications such as a P2P payment system.

This study of this project aims to bridge this gap and show the possibility and feasibility of the offline physical vibration-based P2P payment system. This research will not only demonstrate the design of the payment system and the algorithm of the vibration communication but also focus on developing and evaluating the cryptographic framework to secure it against a defined or possible attack, thereby providing a foundation for secure data transfer over a physical sensory channel.

2. Problem statement, project objectives & scope

The current research on vibration-based P2P communication shows that the method offers better physical security. Since it is a radio-free data exchange method between the mobile phone devices, it makes it impossible for usual wireless attacks to happen due to the close physical contact for communication of vibration signals, which enhances the physical security for authentication and pairing (Hwang et al., 2014). However, the core problem is that the use of high-overhead traditional cryptographic protocols is not suitable in the case of vibration-based communication with severe constraints. As Kim et al. (2015) found, asymmetric encryption and Public-Key Infrastructure (PKI) are particularly unsuitable. Their computational complexity drains resources, and a significant communication overhead is difficult for a low-bit-rate channel. This creates an important research gap in establishing and designing a lightweight encryption and key exchange mechanism for high-latency, low-throughput vibration-based communication.

Furthermore, as mentioned in the motivation section, the increase in mobile phone users has increased the demand for digital financial services like a P2P payment system for transactions (Belanche et al., 2022). However, most P2P payment systems use wireless technology (Gao et al., 2005). There is also an interesting perspective on developing an offline physical vibration communication for payment that may create innovative methods for physical P2P communication in future research.

This project aims to tackle the insufficiency of the use of the vibration channel in the offline P2P payment system and design a lightweight encryption and key exchange mechanism for this communication channel. The primary deliverables of this project shall align with the following objectives:

- To design a Vibration-Based Communication Protocol for the system: Develop an application-based algorithm for reliably encoding, transmitting, and decoding the data between two Android smartphones using the devices' vibrator and accelerometer.
- To develop a Lightweight Cryptographic Framework: Research and implement a simple security mechanism for the vibration channel. These include a secure custom method for secret key exchange and a simple encryption scheme for the transaction data.
- To implement a functional P2P payment System: Develop an Android application prototype demonstrating a secure P2P Transaction system using the proposed communication protocol and cryptographic framework.
- Evaluation of the system performance and security: Test the final prototype of the application, and evaluate its performance, such as data rate, bit error rate, or latency. Moreover, analyze its security strength against a defined possible threat model, and identify any possible vulnerabilities and limitations.

Below are the scope and constraints of the project:

- The project is scoped to Android smartphones due to the hardware consistency and development accessibility (It supports more libraries, and it has more SDK Flexibility)
- The communication is strictly device-to-device via physical contact. It will not involve any other methodology such as Wi-Fi, Bluetooth, or NFC.
- The cryptographic and vibration channel design will be software-based, using the modular libraries provided by different APIs and utilizing existing device hardware (vibrator, accelerometer).
- Performance evaluation will first be conducted in a controlled environment, such as putting it in a table and testing it on two device models.

Below are the assumptions of the project:

- Mobile devices should be equipped with a functional vibrator motor and an accelerometer.
- Users can physically place their devices in contact (e.g., back-to-back) for the duration of the transaction.
- Making the research easier, the project assumes an attacker does not have physical possession of either user's device prior to the payment.
- There are only two users (Device) of the system, and the default currency of the transaction should be HKD.

Below are some cases that will not be considered during the project:

- The vibration algorithm and security analysis will be practical rather than based on formal mathematical proofs (However, there will be some mathematical proofs in the final paper to show how the encryption works in the vibration channel).
- The project will not be a large-scale development but will only focus on the transaction of two different users.
- The possible threat model will only focus on simple scenarios, like eavesdropping or replay attacks. It will not consider complex cases, like injecting the vibration at an extremely close range, which can cause interference.
- Development on IOS or other OS is not within the project scope.
- The final application testing will not be conducted in a highly destructive environment, but in a controllable environment for better results.

3. Major technical components

For the overview of the System to be implemented, this system is a P2P Android application that facilitates a secure payment transaction between two Android mobile devices using vibration for data transmission and a custom cryptographic protocol for security. This process involves a handshake, a secure key exchange over the vibration channel. The following shows an overview of the proposed system.

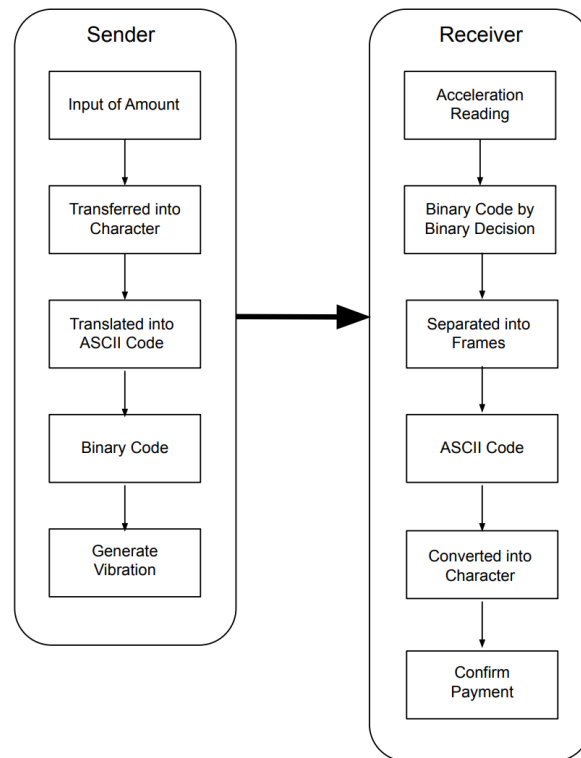


Fig.1. An overview of the proposed system (Tentative)

Below are some short descriptions of each possible technical component of the proposed system. All the components will be developed based on the Kotlin programming Language with the Android Studio IDK and the API support from Android Developers.

1. User Interface (UI) component:

It provides an interaction menu for the users. It includes:

- Login and Registration screen: Users can log in and register accounts for the payment system.
- Menu & Transaction Screen: To display the account balance, input the payment amount, initiate the send request, and listen to the vibration pattern to receive payment.
- Confirmation Screen displays the transaction details.
- Status Indicators: Visual Feedback like “Connecting...”, “Sending payment...”, “Transaction Complete”.

2. Vibration Controller component: A low-level component controls the device’s vibrator start and stop with a specified pattern based on the Encoder (API: *VibrationEffect*).

3. Accelerometer Monitor component: This component is responsible for getting the raw sensor data and a context detector for passing it to the vibration decoder for processing. This

component uses the *SensorManager* API to obtain the (x, y, z) acceleration data using the *TYPE_ACCELEROMETER* sensor.

An overview of the design of vibration communication is presented below. It seeks to combine a range of scholarly approaches into a new algorithm, informed by the contributions of Hwang et al. (2012, 2014) and Yonezawa et al. (2011, 2014). It is separated into the Vibration Encoder and the Vibration Decoder.

4. Vibrator Encoder (Transmitter Design) component:

It translates the payment commands (Single characters) from the user input and packages them for physical transmission according to the super frame structure. The following section will show how the component may work.

- a. The encoder first receives the payment command character like “a”, “b”, or “c” (where a = \$100, b = \$200, c = \$300), based on the transmitter’s selection of the transaction amount to the receiver.
- b. Second, it will convert the character to its 7-bit ASCII Code.
- c. The next step is constructing each character's super frame (Total: 11 frames). The super frame is constructed into an array with the structure below.
 - (a) Beacon Bit (1 frame): It is a bit to announce the start of the super frame of each character and allow the receiver to synchronize it.
 - (b) Active Frames (7 frames): These frames contain an ASCII code converted into a sequence of binary bits. In the binary bits, “1” represents the vibration that has been generated. A “0” means there is no vibration.
 - (c) Inactive Frames (3 frames): A set of frames between the end of each super frame and the start of the next super frame. The design of these three frames provides a guard period to prevent echoes from the signal reflected from the boundaries and to secure time synchronization of each super frame.

The time of each frame should be one second long to keep the communication reliable. The framework output should be like *[beacon, bit1, bit2, ..., bit7, inactive, inactive, inactive]* for the vibration controller to execute.

5. Vibration Decoder (Receiver Design) component:

This component has a sliding window and an adaptive threshold algorithm to look for the start/stop of deliberate vibrations and decode the vibration patterns back to the payment character command. The following section will show how the component may work for the decoder.

- a. First, the decoder looks for the distinctive pattern of the beacon bit, so it knows this is the start of the super frame of each character.

- b. For each 1 second of frame, it will calculate the acceleration data based on the adaptive threshold algorithm to decide the binary (1 or 0) of each frame.
- c. Next, the seven active bits will be reassembled from the super frame and converted to a binary number back to an ASCII character.
- d. Finally, once a character is received, the decoded character (e.g., a = \$100) is immediately processed in the application, triggering the corresponding payment transaction.

6. Cryptographic Protocol component:

This component involves a simple shared key exchange before the transaction and exchanging encrypted data. Due to the time constraint problem for the final year project, the key exchange process will be simplified to a passphrase-based process, but will not use vibration. Below, we briefly show how the cryptographic framework may work.

- a. First, an initial one-time setup is required for each transaction so that both devices can derive a symmetric key from the same passphrase entered by the users in the app.
- b. Second, the system will feed the passphrase into a Key Derivation Function (KDF). The system will generate a fixed salt for both devices and hash the passphrase for many iterations. Finally, KDF will output a strong Pre-Shared Secret key (PSS key) and store it in the Android Keystore for each device.
- c. Third, after the sender and receiver have obtained the PSS key, they will start the vibration payment transaction.
 - a. First, the sender will generate a random 128-bit session key and encrypt the payment data with the session key using AES-128 to get the ciphertext payment data. Next, the session key will be encrypted using the PSS key to get the encrypted secret key. Then, it will package the encrypted secret key and ciphertext payment data into a vibration super frame and send it to the receiver.
 - b. After receiving the vibration signal, the receiver will decode the vibration signal and decrypt the encrypted key with the PSS key to get the session key. Finally, the ciphertext payment data will be decrypted with the session key to get the detailed payment data.

4. Expected results & deliverables

The system will be tested on two Android phones, the Samsung Galaxy A8+ and the Huawei P30 Pro, in a controlled environment, such as on a table. The testing process will include functional testing to ensure the application can complete a full payment transaction use case. Moreover, performance testing will be conducted to investigate the package delivery rate and the total time required for the whole

process (Key exchange and transaction completion). Lastly, security testing will be performed to verify that each identical transaction process will produce a different ciphertext result due to the key exchange.

The expected outcome is to develop a functional, end-to-end secure P2P payment prototype that operates using vibration only. Ideally, a successful transaction should take between 120 and 180 seconds, with a package delivery rate exceeding 90%. The key exchange process is anticipated to be the most time-consuming aspect. This project aims to demonstrate that a secure cryptographic protocol can be effectively implemented over a low-bandwidth rate and high-latency physical channel and prove that using vibration channels in an offline P2P payment system is theoretically workable.

The final deliverables of this FYP will be shown below:

- A complete Android Application (APK): The working prototype app implementing the entire payment system, including the user interface, vibration communication, and cryptographic framework.
- FYP Documentation: A final report that outlines the proposed system design, methodology, system implementation, results, and discussion
- Demonstration: A demo video briefly introduces the project and shows a successful transaction between the two test phones.

5. Project schedule

The project follows an Agile software development approach, with iterative development cycles, which allows higher flexibility. The key milestones and schedule are outlined below:

a) Phase 1: P2P Payment System Interface (September 2025)

- **Goals:** Develop a P2P payment system interface for two Android phones.
- **Key Deliverables:** A functional payment interface with basic transaction operations.

b) Phase 2: Basic Vibration channel development (October – November 2025)

- **Goals:** Set up the development environment and implement core vibration transmission and reception between two phones.
- **Key Deliverables:** A basic Android app that can encode a string into a vibration super frame and decode it back into a string on the receiver device.

c) Phase 3: Cryptographic framework development & Integration (December 2025 – January 2026)

- **Goals:** Design and develop the passphrase-based key exchange and AES encryption/decryption framework and integrate it into the vibration communication layer.
- **Key Deliverables:** An Android app prototype that can perform a secure key exchange and send an encrypted message via vibration.

d) *Phase 4: System Integration & Testing (February 2026)*

- **Goals:** Integrate all components into one complete system and conduct functional, performance, and security testing on testing devices (Samsung A8+, Huawei P30 Pro).
- **Key Deliverables:** A fully functional end-to-end secure payment prototype and mark down the test result detailing the package delivery rate and bitwise accuracy based on different environments and the adaptive threshold algorithm.

e) *Phase 5: Final Preparation & Documentation (March 2026)*

- **Goals:** Prepare the final report, create a demo video, and fine-tune the source code for submission.
- **Key Deliverables:** Final project report, complete source code, and a demonstration video.

Below is the Gantt Chart for the Final Year project (Divided into documentation & System development Gantt Chart):

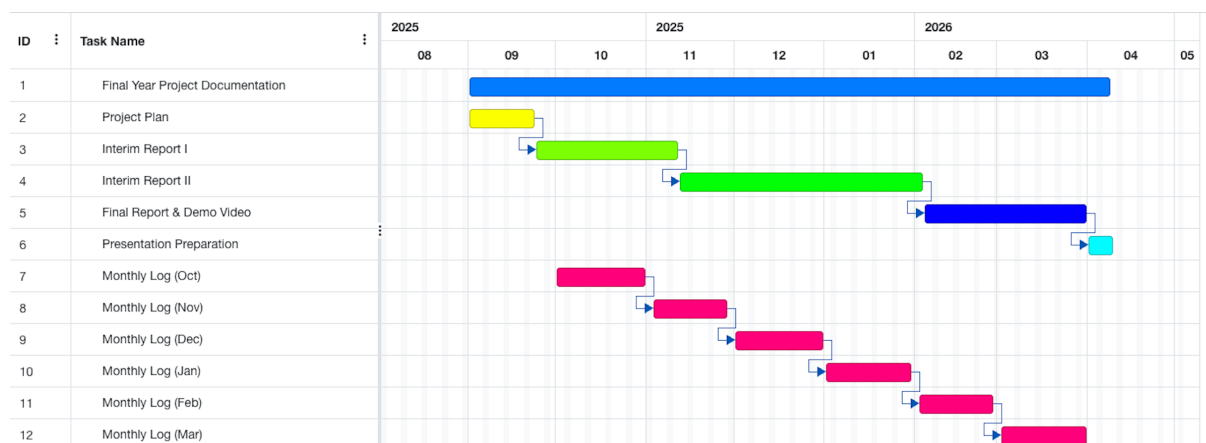


Fig.2. A Gantt Chart for the Documentation part of the Project

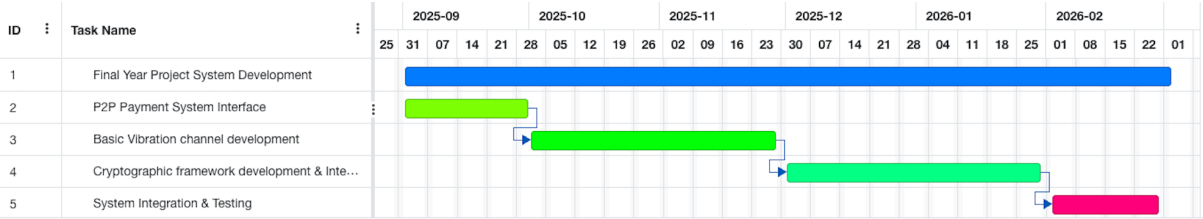


Fig.3. A Gantt Chart for the System Development part of the Project

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